

Qatar-2b

R-1037

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160525 · created 2026-07-30T07:18:56+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457533.67009 +/- 0.00028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.039 +/- 0.02
Transit depth (Rp/Rs)^2	0.15 +/- 0.16 [%]
Orbital Inclination (inc)	89.09 +/- 1.73
Airmass coefficient 1 (a1)	1.117 +/- 0.016
Airmass coefficient 2 (a2)	-0.082 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	None
Optimal Aperture	3.79
Optimal Annulus	5.8
Transit Duration (day)	0.0457 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.039 ± 0.02 vs 0.16327 (deviation 5.75σ)
Duration fit vs published	0.0457 d vs 0.0767 d (deviation 20.48σ)
Mid-time O–C	0.8 min
Depth SNR	1.8
Residual scatter	1.47 %

Light curve

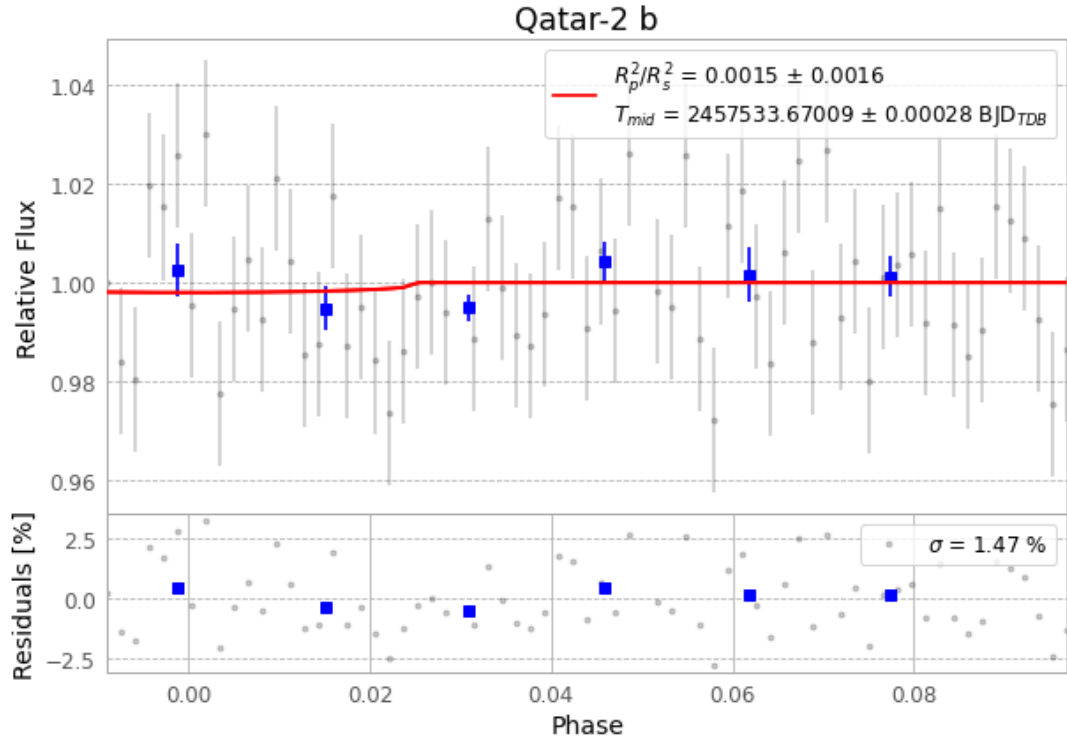


Figure 1 — FinalLightCurve_Qatar-2 b_2016-05-24.png (SHA-256 382ef6da030925c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 210], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b00cc1ac9cfb3c36...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 77e40fd

Data provenance

68 FITS frames (45 MB), Qatar-2160525033912.FITS → Qatar-2160525070312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160525.log (07253e47af85...), FinalLightCurve_Qatar-2 b_2016-05-24.png (382ef6da0309...), FinalLightCurve_Qatar-2 b_2016-05-24.csv (8c2ae228e3c7...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 07:18Z.